	B.M.S. COLLEGE OF ENGINEERING, BANGALORE-19 (Autonomous Institute, Affiliated to VTU) **Department Name: CSE**

SECOND INTERNALS

Course Code: 23CS6PCMAL	**Course Title:** Machine Learning	
Semester: 6th	**Maximum Marks: 40**	**Date: 07/04/2026**
Faculty Handling the Course:	Rekha G S, Dr.Kayarvizhy N, Dr. Pallavi G B, Saritha A N	
Instructions: ***Internal choice is provided in Part C.***		

PART-A

Total 5 Marks (No choice)

No.	Question	Marks	CO	PO	BL
1a	Explain Linear and Non-Linear Support Vector Machine **Linear SVM (2 marks):** • A Support Vector Machine that separates data using a **straight line (hyperplane)**. • Suitable for **linearly separable data**. • Maximizes the **margin** between classes using support vectors. • Equation: $w^T x+b=0$ **Non-Linear SVM (2 marks):** • Used when data is **not linearly separable**. • Applies **kernel trick** to map data into higher dimensions. • Uses kernels like **RBF, Polynomial** to create **decision boundaries**. **Difference / Conclusion (1 mark):** • Linear SVM → simple, faster, straight boundary. • Non-linear SVM → complex, slower, curved boundary for real-world data.	**5M**	**1**	**1**	**2**

PART-B

Total 15 Marks (No Choice)

No.	Question	Marks	CO	PO	BL
2a	A data scientist is working on a loan default prediction system for a bank. The dataset contains noisy data, class imbalance, and complex feature interactions. The initial model using a single decision tree shows poor performance. The data scientist considers using ensemble techniques. i. Which ensemble technique would the data scientist apply to reduce variance in the model caused by noisy data? Explain how it works in this context. i. To reduce variance caused by noisy data, the data scientist should use **bagging (Bootstrap Aggregating)**, particularly models like a Random Forest.	**5M**	**2**	**2**	**4**

	Bagging works by: • Creating multiple bootstrap samples (random samples with replacement) from the original dataset. • Training a separate decision tree on each sample. • Aggregating predictions (e.g., majority voting for classification). **In this context:** Because the dataset is noisy, a single decision tree may overfit to random fluctuations or noise. Bagging reduces this by averaging across many trees, each trained on slightly different data. This smooths out the noise and lowers variance, leading to more stable and generalizable predictions for loan defaults. ii. Which technique would the data scientist apply to improve performance on misclassified loan defaulters? Describe how the method adjusts learning. To improve performance on misclassified loan defaulters, the data scientist should use **boosting**. Boosting works by: • Training models sequentially rather than independently. • Giving more weight (importance) to incorrectly classified instances in each subsequent model. • Combining all models into a final strong learner. **In this context:** If loan defaulters (especially minority class cases) are being misclassified, boosting will: • Focus more on those hard-to-predict cases by increasing their influence. • Adjust the model to learn complex patterns and feature interactions. • Iteratively reduce errors by correcting mistakes from previous models. This makes boosting especially effective for handling class imbalance and improving prediction accuracy on difficult cases like loan defaulters.				
2b	A student evaluates a classifier using only a single train-test split and reports the result as final performance. Analyze the limitations of this approach and explain how resampling techniques improve reliability of performance estimation. **Limitations of Single Train-Test Split (2Marks):** • **High variance**: Result depends on one random split → may not reflect true performance. • **Overfitting risk**: Model may perform well only on that specific test set. • **Data bias**: Train/test split may not represent overall dataset distribution.	**5M**	**2**	**2**	**4**

	How Resampling Improves Reliability (3Marks): • **Multiple evaluations**: Techniques like **k-fold cross-validation** use different splits → more robust estimate. • **Reduced variance**: Averaging results across folds gives stable performance. Resampling methods give **more accurate, stable, and generalizable performance estimates** than a single split. (1Mark)				
2c	Two classifiers are evaluated using k-fold cross-validation, and their mean accuracies are: Model A: 88% and Model B: 90%. A student concludes that Model B is better. Analyze why this conclusion may be incorrect and explain how a paired t-test helps in making a statistically valid comparison. **Why conclusion may be incorrect (2 Marks):** • **Mean accuracy alone is insufficient**: Does not show variability across folds. • **Small difference (88% vs 90%)** may be due to **random chance**. • **Overlap in performance** across folds may exist → models may not be truly different. • Ignores **statistical significance** of results. **How paired t-test helps (2 Marks):** • Compares **fold-wise performance differences** between Model A and Model B. • Tests whether the **mean difference is statistically significant**. • Considers **variance across folds**, not just average accuracy. • Provides a **p-value** to decide if improvement is real or due to chance. **Conclusion (1 point):** Paired t-test ensures a **reliable and statistically valid comparison** before concluding one model is better.	**5M**	**2**	**2**	**4**

PART- C

Total 20 Marks (Choice)

No.	Question	Marks	CO	PO	BL
3a	A streaming platform wants to recommend movies to users. The model predicts whether a user will like (1) or not like (0) a movie.The system is evaluated on 5,000 user-movie interactions. **Predicted Like** / **Predicted Not Like** **Actual Like**: 400 / 100 **Actual Not Like**: 300 / 4200	10M	**2**	**2**	**4**

i. Using the confusion matrix above, compute Accuracy, Precision,Recall (Sensitivity), F1 Score

Accuracy = 92%
Precision ≈ 0.57 (57.1%)
Recall = 0.80 (80%)
F1 Score ≈ 0.67

ii. What does precision indicate in this recommendation system? And What does recall indicate?

Precision:
Precision tells us: Out of all movies the system recommended (predicted "Like"), how many did users actually like?
→ High precision = fewer irrelevant recommendations.

Recall:
Recall tells us: Out of all movies the user would actually like, how many did the system successfully recommend?
→ High recall = fewer missed good movies.

iii. Which metric should be prioritized to recommend only relevant movies? Which metric should be prioritized to ensure users don't miss good movies? Is accuracy reliable here? Why

- To recommend only relevant movies:

→ Prioritize Precision
Because you want to avoid suggesting movies users won't like.

- To ensure users don't miss good movies:

→ Prioritize Recall
Because missing a good movie (false negative) hurts user satisfaction.

- Is Accuracy reliable here?

→ No, it is misleading.
Because:

- The dataset is imbalanced (many more "Not Like" cases: 4500 vs 500).
- A model could predict mostly "Not Like" and still get high accuracy.
- Here, 92% accuracy hides the fact that precision is only ~57%.

iv. Another recommendation model gives: Precision = 0.75, Recall = 0.60 Which model would you choose for the recommendation system? Justify your answer based on user experience goals.

`Current model: Precision ≈ 0.57 Recall = 0.80`

New model: Precision = 0.75 Recall = 0.60

Choice depends on the goal, but typically for recommendation systems:

- Users dislike irrelevant recommendations more than missing some good ones.
- So precision is often prioritized.

Choose the second model (Precision = 0.75, Recall = 0.60)

OR

3b	A hospital is evaluating two machine learning models (Model A and Model B) for early cancer detection. Since cancer cases are rare but critical, choosing the right model is essential. The performance of the models is summarized below:	**10M**	**2**	**2**	**4**

Metric	Model A	Model B
Accuracy	96%	94%
Precision	0.85	0.70
Recall	0.60	0.85
F1 Score	0.70	0.77

i. Analyze the trade-off between precision and recall for both models. Explain how each model behaves in detecting cancer cases.
ii. Evaluate which model is more suitable for early cancer detection. Justify your choice based on medical priorities and risk factors.
iii. Critically analyze whether accuracy is a reliable metric in this scenario. Support your reasoning considering class imbalance.
iv. Suppose the hospital wants to reduce both false positives and false negatives. Analyze how the F1-score helps in model selection and which model performs better.

i. Precision–Recall Trade-off (2.5 marks)

- **Model A**:
  - High precision (0.85) → **fewer false positives**
  - Low recall (0.60) → **misses many actual cancer cases (high false negatives)**
- **Model B**:
  - Lower precision (0.70) → **more false positives**
  - High recall (0.85) → **detects most cancer cases (low false negatives)**
- **Trade-off**:
  - Model A is **conservative** (avoids false alarms).
  - Model B is **sensitive** (captures more true cancer cases).

ii. Suitable Model for Early Cancer Detection (2.5 marks)

- **Model B is more suitable**
- Reason:
  - In healthcare, **missing a cancer case (false negative) is more dangerous** than a false alarm.
  - High recall (0.85) ensures **early detection and timely treatment**.
- False positives can be further tested, but **false negatives can be fatal**.

iii. Is Accuracy Reliable? (2.5 marks)

- **Accuracy is misleading in imbalanced datasets** (cancer cases are rare).
- A model can achieve high accuracy by predicting mostly **non-cancer cases**.
- Example: Even with 96% accuracy, Model A misses many real cases (low recall).
- Hence, accuracy **does not reflect true performance** for critical minority class.

iv. Role of F1-Score (2.5 marks)

- **F1-score balances precision and recall**.

- Useful when both **false positives and false negatives must be minimized**.
- **Model B has higher F1-score (0.77 vs 0.70)** → better overall balance.
- Indicates Model B provides a **more reliable trade-off**.

Model B is preferred due to **higher recall and better F1-score**, making it safer and more effective for early cancer detection.

OR

4a **10M** **2** **2** **4**

A medical dataset is used to predict disease (Positive/Negative) based on attributes: Fever (Yes/No) ,Cough (Yes/No) Test Result (Positive/Negative)

Fever	Cough	Test Result	Class
Yes	Yes	Positive	Positive
Yes	No	Positive	Positive
No	Yes	Negative	Negative
Yes	Yes	Negative	Positive
No	No	Negative	Negative
No	Yes	Positive	Positive

i. Compute entropy of the dataset.
ii. Calculate information gain for all attributes.
iii. Identify the root node.
iv. Analyze how the distribution of class labels affects the choice of root attribute.

i. Entropy of Dataset (2.5 marks)

- Total samples = 6
- Positive = 4, Negative = 2

$$H(S) = -\left(\frac{4}{6}\log_2\frac{4}{6} + \frac{2}{6}\log_2\frac{2}{6}\right)$$

- $H(S) \approx 0.918$

ii. Information Gain (4 marks)

1. Fever

- Fever = Yes → (3P, 0N) → Entropy = 0
- Fever = No → (1P, 2N) → Entropy ≈ 0.918
- Weighted entropy ≈ 0.459
- **IG(Fever) = 0.918 – 0.459 = 0.459**

2. Cough

- Cough = Yes → (3P, 1N) → Entropy ≈ 0.811
- Cough = No → (1P, 1N) → Entropy = 1
- Weighted entropy ≈ 0.874
- **IG(Cough) = 0.918 – 0.874 = 0.044**

	3. Test Result • Positive → (3P, 0N) → Entropy = 0 • Negative → (1P, 2N) → Entropy ≈ 0.918 • Weighted entropy ≈ 0.459 • **IG(Test Result) = 0.918 – 0.459 = 0.459** iii. Root Node (1.5 marks) • Highest Information Gain = **Fever = 0.459** and **Test Result = 0.459** • **Either can be chosen as root node** (tie case). iv. Effect of Class Distribution (2 marks) • Dataset is **imbalanced (4 Positive, 2 Negative)** → entropy is moderate (0.918). • Attributes that create **pure splits (all positive or all negative)** give higher IG. • Both Fever and Test Result produce **perfectly pure subsets**, hence selected. • Skewed class distribution influences IG by favoring attributes that **separate dominant class clearly**.				
	OR				
4b	A dataset contains the following labeled points: Positively labeled data points: $\{(2,2),(4,2),(4,4)\}$ Negatively labeled data points**:** $\{(0,0),(2,0),(0,2)\}$ From the given dataset, the following support vectors are identified:$S_1 = (2,0), S_2 = (2,2), S_3 = (0,2)$. Using the given support vectors, determine the equation of the optimal separating hyperplane. Further, analyze how the support vectors influence the margin and justify why only these points determine the decision boundary. **Positive class (+1):** (2,2), (4,2), (4,4) **Negative class (−1):** (0,0), (2,0), (0,2) **Support vectors:** S_1 = (2,0) ⟶ −1 S_2 = (2,2) ⟶ +1 S_3 = (0,2) ⟶ −1	**10 M**	**2**	**2**	**4**

Apply to each support vector

1. S_1 = (2,0), label −1

$$2w_1 + 0w_2 + b = -1$$

$$2w_1 + b = -1 \quad ...(1)$$

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2. S_2 = (2,2), label +1

$$2w_1 + 2w_2 + b = 1 \quad ...(2)$$

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3. S_3 = (0,2), label −1

$$0w_1 + 2w_2 + b = -1$$

$$2w_2 + b = -1 \quad ...(3)$$

Solve equations

Final Hyperplane

x+y−3=0

Margin is:

$$\frac{2}{\|w\|} = \frac{2}{\sqrt{1^2 + 1^2}} = \frac{2}{\sqrt{2}} = \sqrt{2}$$

Analysis

Why only these points matter:

- Support vectors lie **closest to the boundary**
- They satisfy:

$w \cdot x + b = \pm 1$ They **define the margin edges**

Influence on margin:

- Margin depends on ‖w‖
- www is computed only from support vectors
- Moving any support vector → boundary shifts
- Moving non-support vectors → no effect

S_1 and S_3 (negative class) form one boundary line

	S_2 (positive class) forms the opposite boundary The optimal hyperplane sits exactly midway b**etween them**				